

03-01-06_InfoSharingBinomialLRT

setup

```
source("../_common.R")
```

libraries

```
library(openxlsx)  
library(afex)
```

Loading required package: lme4

Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

expand, pack, unpack

Welcome to afex. For support visit: <http://afex.singmann.science/>

- Functions for ANOVAs: `aov_car()`, `aov_ez()`, and `aov_4()`
- Methods for calculating p-values with `mixed()`: 'S', 'KR', 'LRT', and 'PB'
- 'afex_aov' and 'mixed' objects can be passed to `emmeans()` for follow-up tests
- Get and set global package options with: `afex_options()`
- Set sum-to-zero contrasts globally: `set_sum_contrasts()`
- For example analyses see: `browseVignettes("afex")`

Attaching package: 'afex'

The following object is masked from 'package:lme4':

`lmer`

```
library(lme4)
library(emmeans)
```

Welcome to emmeans.

Caution: You lose important information if you filter this package's results.
See '? untidy'

```
library(papaja)
```

Loading required package: tinylab

```
# library(glmmTMB)
# library(modelsummary)
# library(gtsummary)
# library(broom)
# library(DHARMa)

# library(RColorBrewer)
# library(tibble)
```

read data

```
dat_long <- read.xlsx("../artifacts/binominal_data.xlsx")
```

```
sapply(
  dat_long %>%
    select(
      Q14.1,
      sex,
      SES_mean,
      hispanic,
      NeedCog_Mean,
      DirtyDozen_Narcissism,
      DirtyDozen_Psychopathy,
      DirtyDozen_Machiavellianism,
      SVO_angle
    ),
  class
)
```

Q14.1	sex
"numeric"	"numeric"
SES_mean	hispanic
"numeric"	"numeric"
NeedCog_Mean	DirtyDozen_Narcissism
"numeric"	"numeric"
DirtyDozen_Psychopathy	DirtyDozen_Machiavellianism
"character"	"numeric"
SVO_angle	
"numeric"	

```
dat_long <- dat_long %>%
  mutate(
    DirtyDozen_Psychopathy = as.numeric(DirtyDozen_Psychopathy)
  )
```

update data

Remove any cases which are missing. Causing warnings/issues with the model

```
control_data <- dat_long %>%
  select(
    ItemResponse,
    InfoDistribution,
    InfoImportance,
    Q14.1,
    sex,
    SES_mean,
    hispanic,
    NeedCog_Mean,
    DirtyDozen_Narcissism,
    DirtyDozen_Psychopathy,
    DirtyDozen_Machiavellianism,
    SVO_angle,
    ResponseId,
    ItemID
  ) %>%
  drop_na() %>%
  droplevels()
```

Center variables to support model convergence

```
control_data <- control_data %>%
  mutate(
    Age_z = as.numeric(scale(Q14.1)),
    SES_z = as.numeric(scale(SES_mean)),
    NFC_z = as.numeric(scale(NeedCog_Mean)),
    Narcissism_z = as.numeric(scale(DirtyDozen_Narcissism)),
    Psychopathy_z = as.numeric(scale(DirtyDozen_Psychopathy)),
    Machiavellianism_z = as.numeric(scale(DirtyDozen_Machiavellianism)),
    SVO_z = as.numeric(scale(SVO_angle))
  )
```

control model

```
model_binom_lrt_controls <- mixed(
  ItemResponse ~
    InfoDistribution +
    InfoImportance +
    Age_z +
```

```

sex +
SES_z +
hispanic +
NFC_z +
Narcissism_z +
Psychopathy_z +
Machiavellianism_z +
SVO_z +
(1 | ResponseId) +
(1 | ItemID),
data = control_data,
family = binomial(link = "logit"),
method = "LRT",
control = glmerControl(
  optimizer = "bobyqa",
  optCtrl = list(maxfun = 200000)
)
)

```

Contrasts set to `contr.sum` for the following variables: InfoDistribution, InfoImportance, Res

Numerical variables NOT centered on 0: sex, hispanic

If in interactions, interpretation of lower order (e.g., main) effects difficult.

`model_binom_lrt_controls`

Mixed Model Anova Table (Type 3 tests, LRT-method)

Model: ItemResponse ~ InfoDistribution + InfoImportance + Age_z + sex +

Model: SES_z + hispanic + NFC_z + Narcissism_z + Psychopathy_z +

Model: Machiavellianism_z + SVO_z + (1 | ResponseId) + (1 | ItemID)

Data: control_data

Df full model: 14

	Effect	df	Chisq	p.value
1	InfoDistribution	1	3.34 +	.068
2	InfoImportance	1	10.28 **	.001
3	Age_z	1	0.82	.364
4	sex	1	5.80 *	.016
5	SES_z	1	0.39	.530
6	hispanic	1	0.07	.797
7	NFC_z	1	10.02 **	.002

8	Narcissism_z	1	1.25	.263
9	Psychopathy_z	1	0.47	.494
10	Machiavellianism_z	1	0.80	.372
11	SV0_z	1	0.00	.988

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '+' 0.1 ' ' 1

Only sex and NFC were significant

remove old data

```
rm(control_data)
```

full model

create data

```
full_model_data <- dat_long %>%
  select(
    ItemResponse,
    structure_number01,
    motivation_number01,
    InfoDistribution,
    InfoImportance,
    sex,
    NeedCog_Mean,
    ResponseId,
    ItemID
  ) %>%
  drop_na() %>%
  droplevels() %>%
  mutate(
    NeedCog_z = as.numeric(scale(NeedCog_Mean))
  )
```

model

Model includes optimization and centering of variables because of excessive warnings and failure to converge in previous attempts.

```

model_binom_lrt <- mixed(
  ItemResponse ~
    structure_number01 *
    motivation_number01 *
    InfoDistribution *
    InfoImportance +
    sex +
    NeedCog_z +
    (1 | ResponseId) +
    (1 | ItemID),
  data = full_model_data,
  family = binomial(link = "logit"),
  method = "LRT",
  control = glmerControl(
    optimizer = "bobyqa",
    optCtrl = list(maxfun = 200000)
  )
)

```

Contrasts set to contr.sum for the following variables: structure_number01, motivation_number01

Numerical variables NOT centered on 0: sex

If in interactions, interpretation of lower order (e.g., main) effects difficult.

```
model_binom_lrt
```

Mixed Model Anova Table (Type 3 tests, LRT-method)

Model: ItemResponse ~ structure_number01 * motivation_number01 * InfoDistribution *

Model: InfoImportance + sex + NeedCog_z + (1 | ResponseId) + (1 |

Model: ItemID)

Data: full_model_data

Df full model: 20

	Effect	df
1	structure_number01	1
2	motivation_number01	1
3	InfoDistribution	1
4	InfoImportance	1
5	sex	1
6	NeedCog_z	1
7	structure_number01:motivation_number01	1

8	structure_number01:InfoDistribution	1
9	motivation_number01:InfoDistribution	1
10	structure_number01:InfoImportance	1
11	motivation_number01:InfoImportance	1
12	InfoDistribution:InfoImportance	1
13	structure_number01:motivation_number01:InfoDistribution	1
14	structure_number01:motivation_number01:InfoImportance	1
15	structure_number01:InfoDistribution:InfoImportance	1
16	motivation_number01:InfoDistribution:InfoImportance	1
17	structure_number01:motivation_number01:InfoDistribution:InfoImportance	1

	Chisq	p.value
1	0.30	.586
2	15.29 ***	<.001
3	3.32 +	.068
4	10.47 **	.001
5	7.25 **	.007
6	15.04 ***	<.001
7	4.10 *	.043
8	3.56 +	.059
9	0.06	.808
10	6.84 **	.009
11	4.29 *	.038
12	0.13	.722
13	0.06	.809
14	11.48 ***	<.001
15	1.33	.249
16	0.13	.718
17	12.26 ***	<.001

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '+' 0.1 ' ' 1

```
anova(model_binom_lrt)
```

Mixed Model Anova Table (Type 3 tests, LRT-method)

Model: ItemResponse ~ structure_number01 * motivation_number01 * InfoDistribution *
Model: InfoImportance + sex + NeedCog_z + (1 | ResponseId) + (1 |
Model: ItemID)

Data: full_model_data

Df full model: 20

	Df
structure_number01	19

motivation_number01	19
InfoDistribution	19
InfoImportance	19
sex	19
NeedCog_z	19
structure_number01:motivation_number01	19
structure_number01:InfoDistribution	19
motivation_number01:InfoDistribution	19
structure_number01:InfoImportance	19
motivation_number01:InfoImportance	19
InfoDistribution:InfoImportance	19
structure_number01:motivation_number01:InfoDistribution	19
structure_number01:motivation_number01:InfoImportance	19
structure_number01:InfoDistribution:InfoImportance	19
motivation_number01:InfoDistribution:InfoImportance	19
structure_number01:motivation_number01:InfoDistribution:InfoImportance	19
	Chisq
structure_number01	0.2966
motivation_number01	15.2929
InfoDistribution	3.3247
InfoImportance	10.4716
sex	7.2462
NeedCog_z	15.0422
structure_number01:motivation_number01	4.1008
structure_number01:InfoDistribution	3.5609
motivation_number01:InfoDistribution	0.0588
structure_number01:InfoImportance	6.8430
motivation_number01:InfoImportance	4.2928
InfoDistribution:InfoImportance	0.1263
structure_number01:motivation_number01:InfoDistribution	0.0582
structure_number01:motivation_number01:InfoImportance	11.4819
structure_number01:InfoDistribution:InfoImportance	1.3279
motivation_number01:InfoDistribution:InfoImportance	0.1302
structure_number01:motivation_number01:InfoDistribution:InfoImportance	12.2557
	Chi Df
structure_number01	1
motivation_number01	1
InfoDistribution	1
InfoImportance	1
sex	1
NeedCog_z	1
structure_number01:motivation_number01	1
structure_number01:InfoDistribution	1

motivation_number01:InfoDistribution	1
structure_number01:InfoImportance	1
motivation_number01:InfoImportance	1
InfoDistribution:InfoImportance	1
structure_number01:motivation_number01:InfoDistribution	1
structure_number01:motivation_number01:InfoImportance	1
structure_number01:InfoDistribution:InfoImportance	1
motivation_number01:InfoDistribution:InfoImportance	1
structure_number01:motivation_number01:InfoDistribution:InfoImportance	1
	Pr(>Chisq)
structure_number01	0.5860174
motivation_number01	9.206e-05
InfoDistribution	0.0682452
InfoImportance	0.0012123
sex	0.0071050
NeedCog_z	0.0001051
structure_number01:motivation_number01	0.0428627
structure_number01:InfoDistribution	0.0591570
motivation_number01:InfoDistribution	0.8083932
structure_number01:InfoImportance	0.0088989
motivation_number01:InfoImportance	0.0382747
InfoDistribution:InfoImportance	0.7223058
structure_number01:motivation_number01:InfoDistribution	0.8093389
structure_number01:motivation_number01:InfoImportance	0.0007028
structure_number01:InfoDistribution:InfoImportance	0.2491785
motivation_number01:InfoDistribution:InfoImportance	0.7181772
structure_number01:motivation_number01:InfoDistribution:InfoImportance	0.0004638
structure_number01	
motivation_number01	***
InfoDistribution	.
InfoImportance	**
sex	**
NeedCog_z	***
structure_number01:motivation_number01	*
structure_number01:InfoDistribution	.
motivation_number01:InfoDistribution	
structure_number01:InfoImportance	**
motivation_number01:InfoImportance	*
InfoDistribution:InfoImportance	
structure_number01:motivation_number01:InfoDistribution	
structure_number01:motivation_number01:InfoImportance	***
structure_number01:InfoDistribution:InfoImportance	

```

motivation_number01:InfoDistribution:InfoImportance
structure_number01:motivation_number01:InfoDistribution:InfoImportance ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

four panel comparisons

high, common

```

lrt_high_common <- mixed(
  ItemResponse ~
    structure_number01 * motivation_number01 +
    sex +
    NeedCog_z +
    (1 | ResponseId) +
    (1 | ItemID),
  data = full_model_data %>%
    filter(
      InfoImportance == "High Importance",
      InfoDistribution == "Common"
    ),
  family = binomial(link = "logit"),
  method = "LRT",
  control = glmerControl(
    optimizer = "bobyqa",
    optCtrl = list(maxfun = 200000)
  )
)

```

Contrasts set to contr.sum for the following variables: structure_number01, motivation_number01

Numerical variables NOT centered on 0: sex

If in interactions, interpretation of lower order (e.g., main) effects difficult.

```
lrt_high_common
```

Mixed Model Anova Table (Type 3 tests, LRT-method)

```

Model: ItemResponse ~ structure_number01 * motivation_number01 + sex +
Model:      NeedCog_z + (1 | ResponseId) + (1 | ItemID)

```

```
Data: full_model_data %>% filter(InfoImportance == "High Importance",
Data:      InfoDistribution == "Common")
```

```
Df full model: 8
```

	Effect	df	Chisq	p.value
1	structure_number01	1	0.00	.999
2	motivation_number01	1	10.39 **	.001
3	sex	1	10.02 **	.002
4	NeedCog_z	1	7.54 **	.006
5	structure_number01:motivation_number01	1	14.51 ***	<.001

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '+' 0.1 ' ' 1
```

high, unique

```
lrt_high_unique <- mixed(
  ItemResponse ~
    structure_number01 * motivation_number01 +
    sex +
    NeedCog_z +
    (1 | ResponseId) +
    (1 | ItemID),
  data = full_model_data %>%
    filter(
      InfoImportance == "High Importance",
      InfoDistribution == "Unique"
    ),
  family = binomial(link = "logit"),
  method = "LRT",
  control = glmerControl(
    optimizer = "bobyqa",
    optCtrl = list(maxfun = 200000)
  )
)
```

Contrasts set to contr.sum for the following variables: structure_number01, motivation_number01

Numerical variables NOT centered on 0: sex

If in interactions, interpretation of lower order (e.g., main) effects difficult.

```
lrt_high_unique
```

Mixed Model Anova Table (Type 3 tests, LRT-method)

```
Model: ItemResponse ~ structure_number01 * motivation_number01 + sex +
Model:      NeedCog_z + (1 | ResponseId) + (1 | ItemID)
Data: full_model_data %>% filter(InfoImportance == "High Importance",
Data:      InfoDistribution == "Unique")
Df full model: 8
```

	Effect	df	Chisq	p.value
1	structure_number01	1	7.83 **	.005
2	motivation_number01	1	8.57 **	.003
3	sex	1	5.06 *	.025
4	NeedCog_z	1	0.04	.839
5	structure_number01:motivation_number01	1	1.01	.314

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '+' 0.1 ' ' 1

low, common

```
lrt_low_common <- mixed(
  ItemResponse ~
    structure_number01 * motivation_number01 +
    sex +
    NeedCog_z +
    (1 | ResponseId) +
    (1 | ItemID),
  data = full_model_data %>%
    filter(
      InfoImportance == "Low Importance",
      InfoDistribution == "Common"
    ),
  family = binomial(link = "logit"),
  method = "LRT",
  control = glmerControl(
    optimizer = "bobyqa",
    optCtrl = list(maxfun = 200000)
  )
)
```

Contrasts set to contr.sum for the following variables: structure_number01, motivation_number01

Numerical variables NOT centered on 0: sex

If in interactions, interpretation of lower order (e.g., main) effects difficult.

```
lrt_low_common
```

Mixed Model Anova Table (Type 3 tests, LRT-method)

```
Model: ItemResponse ~ structure_number01 * motivation_number01 + sex +
```

```
Model:      NeedCog_z + (1 | ResponseId) + (1 | ItemID)
```

```
Data: full_model_data %>% filter(InfoImportance == "Low Importance",
```

```
Data:      InfoDistribution == "Common")
```

```
Df full model: 8
```

	Effect	df	Chisq	p.value
1	structure_number01	1	0.91	.340
2	motivation_number01	1	1.04	.308
3	sex	1	0.01	.924
4	NeedCog_z	1	7.54 **	.006
5	structure_number01:motivation_number01	1	1.35	.245

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '+' 0.1 ' ' 1
```

low, unique

```
lrt_low_unique <- mixed(  
  ItemResponse ~  
    structure_number01 * motivation_number01 +  
    sex +  
    NeedCog_z +  
    (1 | ResponseId) +  
    (1 | ItemID),  
  data = full_model_data %>%  
    filter(  
      InfoImportance == "Low Importance",  
      InfoDistribution == "Unique"  
    ),  
  family = binomial(link = "logit"),  
  method = "LRT",  
  control = glmerControl(  
    optimizer = "bobyqa",  
    optCtrl = list(maxfun = 200000)  
  )  
)
```

Contrasts set to `contr.sum` for the following variables: `structure_number01`, `motivation_number01`

Numerical variables NOT centered on 0: `sex`

If in interactions, interpretation of lower order (e.g., main) effects difficult.

```
lrt_low_unique
```

Mixed Model Anova Table (Type 3 tests, LRT-method)

```
Model: ItemResponse ~ structure_number01 * motivation_number01 + sex +
```

```
Model:      NeedCog_z + (1 | ResponseId) + (1 | ItemID)
```

```
Data: full_model_data %>% filter(InfoImportance == "Low Importance",
```

```
Data:      InfoDistribution == "Unique")
```

```
Df full model: 8
```

	Effect	df	Chisq	p.value
1	structure_number01	1	0.39	.531
2	motivation_number01	1	3.57 +	.059
3	sex	1	0.79	.373
4	NeedCog_z	1	11.85 ***	<.001
5	structure_number01:motivation_number01	1	1.51	.219

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '+' 0.1 ' ' 1

emmeans

estimation of sharing

emm

```
lc_emm <- emmeans(  
  lrt_low_common,  
  ~ structure_number01 * motivation_number01,  
  type = "response"  
)  
  
hc_emm <- emmeans(  
  lrt_high_common,  
  ~ structure_number01 * motivation_number01,  
  type = "response"  
)
```

```
lu_emm <- emmeans(
  lrt_low_unique,
  ~ structure_number01 * motivation_number01,
  type = "response"
)
```

```
hu_emm <- emmeans(
  lrt_high_unique,
  ~ structure_number01 * motivation_number01,
  type = "response"
)
```

```
hc_structure <- emmeans(
  lrt_high_common,
  ~ structure_number01 | motivation_number01,
  type = "response"
)
```

```
hc_structure
```

```
motivation_number01 = Proself:
  structure_number01  prob  SE  df asymp.LCL asymp.UCL
Intergroup           0.663 0.1240 Inf    0.3989    0.854
Interindividual      0.238 0.0992 Inf    0.0965    0.477
```

```
motivation_number01 = Prosocial:
  structure_number01  prob  SE  df asymp.LCL asymp.UCL
Intergroup           0.603 0.1270 Inf    0.3491    0.811
Interindividual      0.905 0.0512 Inf    0.7476    0.969
```

Results are averaged over the levels of: sex
 Confidence level used: 0.95
 Intervals are back-transformed from the logit scale

```
# pairs(hc_structure, adjust = "holm")
```

```
hc_motivation <- emmeans(
  lrt_high_common,
  ~ motivation_number01 | structure_number01,
  type = "response"
)
```

```
)
```

```
hc_motivation
```

```
structure_number01 = Intergroup:  
motivation_number01  prob      SE  df asymp.LCL asymp.UCL  
Proself              0.663 0.1240 Inf    0.3989    0.854  
Prosocial            0.603 0.1270 Inf    0.3491    0.811
```

```
structure_number01 = Interindividual:  
motivation_number01  prob      SE  df asymp.LCL asymp.UCL  
Proself              0.238 0.0992 Inf    0.0965    0.477  
Prosocial            0.905 0.0512 Inf    0.7476    0.969
```

Results are averaged over the levels of: sex
Confidence level used: 0.95
Intervals are back-transformed from the logit scale

```
# pairs(hc_motivation, adjust = "holm")
```

high, unique

```
hu_structure <- emmeans(  
  lrt_high_unique,  
  ~ structure_number01,  
  type = "response"  
)
```

NOTE: Results may be misleading due to involvement in interactions

```
hu_structure
```

```
structure_number01  prob      SE  df asymp.LCL asymp.UCL  
Intergroup          0.332 0.0820 Inf    0.194    0.507  
Interindividual     0.623 0.0857 Inf    0.447    0.771
```

Results are averaged over the levels of: motivation_number01, sex
Confidence level used: 0.95
Intervals are back-transformed from the logit scale

```
# pairs(hu_structure)
```

```
hu_motivation <- emmeans(  
  lrt_high_unique,  
  ~ motivation_number01,  
  type = "response"  
)
```

NOTE: Results may be misleading due to involvement in interactions

```
hu_motivation
```

motivation_number01	prob	SE	df	asyp.LCL	asyp.UCL
Proself	0.326	0.0805	Inf	0.190	0.498
Prosocial	0.630	0.0865	Inf	0.451	0.779

Results are averaged over the levels of: structure_number01, sex

Confidence level used: 0.95

Intervals are back-transformed from the logit scale

```
# pairs(hu_motivation)
```

LRTs

high, common

structure among proself

Among Proself participants, does Interindividual differ from Intergroup?

```
lrt_hc_structure_proself <- mixed(  
  ItemResponse ~  
    structure_number01 +  
    sex +  
    NeedCog_z +  
    (1 | ResponseId) +  
    (1 | ItemID),  
  data = full_model_data %>%  
    filter(  
      structure_number01 == "Proself"    )
```

```

      InfoImportance == "High Importance",
      InfoDistribution == "Common",
      motivation_number01 == "Proself"
    ),
    family = binomial(link = "logit"),
    method = "LRT",
    control = glmerControl(
      optimizer = "bobyqa",
      optCtrl = list(maxfun = 200000)
    )
  )
)

```

Contrasts set to `contr.sum` for the following variables: `structure_number01`, `ResponseId`, `ItemID`

Numerical variables NOT centered on 0: `sex`, `NeedCog_z`

If in interactions, interpretation of lower order (e.g., main) effects difficult.

```
lrt_hc_structure_proself
```

Mixed Model Anova Table (Type 3 tests, LRT-method)

Model: `ItemResponse ~ structure_number01 + sex + NeedCog_z + (1 | ResponseId) +`

Model: `(1 | ItemID)`

Data: `full_model_data %>% filter(InfoImportance == "High Importance",`

Data: `InfoDistribution == "Common", motivation_number01 == "Proself")`

Df full model: 6

	Effect	df	Chisq	p.value
1	structure_number01	1	4.99 *	.025
2	sex	1	6.61 *	.010
3	NeedCog_z	1	0.04	.835

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '+' 0.1 ' ' 1

structure among prosocial

Among Prosocial participants, does Interindividual differ from Intergroup?

```

lrt_hc_structure_prosocial <- mixed(
  ItemResponse ~
    structure_number01 +

```

```

sex +
NeedCog_z +
(1 | ResponseId) +
(1 | ItemID),
data = full_model_data %>%
  filter(
    InfoImportance == "High Importance",
    InfoDistribution == "Common",
    motivation_number01 == "Prosocial"
  ),
family = binomial(link = "logit"),
method = "LRT",
control = glmerControl(
  optimizer = "bobyqa",
  optCtrl = list(maxfun = 200000)
)
)

```

Contrasts set to contr.sum for the following variables: structure_number01, ResponseId, ItemID

Numerical variables NOT centered on 0: sex, NeedCog_z

If in interactions, interpretation of lower order (e.g., main) effects difficult.

lrt_hc_structure_prosocial

Mixed Model Anova Table (Type 3 tests, LRT-method)

Model: ItemResponse ~ structure_number01 + sex + NeedCog_z + (1 | ResponseId) +

Model: (1 | ItemID)

Data: full_model_data %>% filter(InfoImportance == "High Importance",

Data: InfoDistribution == "Common", motivation_number01 == "Prosocial")

Df full model: 6

	Effect	df	Chisq	p.value
1	structure_number01	1	7.85 **	.005
2	sex	1	6.59 *	.010
3	NeedCog_z	1	13.45 ***	<.001

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '+' 0.1 ' ' 1

motivation within interindividual

Within the Interindividual condition, do Proself and Prosocial participants differ?

```
lrt_hc_motivation_interindividual <- mixed(
  ItemResponse ~
    motivation_number01 +
    sex +
    NeedCog_z +
    # (1 | ItemID) + # Item id removed because its estimated variance was 0, producing singl
    (1 | ResponseId),
  data = full_model_data %>%
  filter(
    InfoImportance == "High Importance",
    InfoDistribution == "Common",
    structure_number01 == "Interindividual"
  ),
  family = binomial(link = "logit"),
  method = "LRT",
  control = glmerControl(
    optimizer = "bobyqa",
    optCtrl = list(maxfun = 200000)
  )
)
```

Contrasts set to contr.sum for the following variables: motivation_number01, ResponseId

Numerical variables NOT centered on 0: sex, NeedCog_z

If in interactions, interpretation of lower order (e.g., main) effects difficult.

```
lrt_hc_motivation_interindividual
```

Mixed Model Anova Table (Type 3 tests, LRT-method)

Model: ItemResponse ~ motivation_number01 + sex + NeedCog_z + (1 | ResponseId)

Data: full_model_data %>% filter(InfoImportance == "High Importance",

Data: InfoDistribution == "Common", structure_number01 == "Interindividual")

Df full model: 5

	Effect	df	Chisq	p.value
1	motivation_number01	1	25.05 ***	<.001
2	sex	1	6.91 **	.009

```
3          NeedCog_z  1          0.72    .396
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '+' 0.1 ' ' 1
```

There is only three items for each of the information types. The output below with Item ID included was showing that the model estimated variance among the items is zero. There was no detectable variation among the items.

```
# VarCorr(lrt_hc_motivation_interindividual$full_model)
```

```
# isSingular(lrt_hc_motivation_interindividual$full_model)
```

motivation within intergroup

Within the Intergroup condition, do Proself and Prosocial participants differ?

```
lrt_hc_motivation_intergroup <- mixed(
  ItemResponse ~
    motivation_number01 +
    sex +
    NeedCog_z +
    (1 | ResponseId) +
    (1 | ItemID),
  data = full_model_data %>%
    filter(
      InfoImportance == "High Importance",
      InfoDistribution == "Common",
      structure_number01 == "Intergroup"
    ),
  family = binomial(link = "logit"),
  method = "LRT",
  control = glmerControl(
    optimizer = "bobyqa",
    optCtrl = list(maxfun = 200000)
  )
)
```

Contrasts set to `contr.sum` for the following variables: `motivation_number01`, `ResponseId`, `ItemID`

Numerical variables NOT centered on 0: `sex`, `NeedCog_z`

If in interactions, interpretation of lower order (e.g., main) effects difficult.

```
lrt_hc_motivation_intergroup
```

Mixed Model Anova Table (Type 3 tests, LRT-method)

```
Model: ItemResponse ~ motivation_number01 + sex + NeedCog_z + (1 | ResponseId) +
Model:      (1 | ItemID)
Data: full_model_data %>% filter(InfoImportance == "High Importance",
Data:      InfoDistribution == "Common", structure_number01 == "Intergroup")
Df full model: 6
      Effect df    Chisq p.value
1 motivation_number01  1    0.11   .737
2              sex    1  3.56 +   .059
3      NeedCog_z    1  9.29 **   .002
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '+' 0.1 ' ' 1
```

check

None of the other models had the issue with ID, so I will keep ID there. It is removed from only the one model because it was the failure point.

```
VarCorr(lrt_hc_structure_proself$full_model)
```

Groups	Name	Std.Dev.
ResponseId	(Intercept)	2.11469
ItemID	(Intercept)	0.29483

```
VarCorr(lrt_hc_structure_prosocial$full_model)
```

Groups	Name	Std.Dev.
ResponseId	(Intercept)	1.24498
ItemID	(Intercept)	0.23743

```
VarCorr(lrt_hc_motivation_interindividual$full_model)
```

Groups	Name	Std.Dev.
ResponseId	(Intercept)	1.1667

```
VarCorr(lrt_hc_motivation_intergroup$full_model)
```

Groups	Name	Std.Dev.
ResponseId	(Intercept)	2.3586
ItemID	(Intercept)	0.5089

high, unique

NA because there were only main effects, no interaction

tables

control variables

coef_table

```
coef_table <- as.data.frame(  
  summary(model_binom_lrt_controls$full_model)$coefficients  
) %>%  
  tibble::rownames_to_column("term") %>%  
  mutate(  
    OR = exp(Estimate),  
    CI_low = exp(Estimate - 1.96 * `Std. Error`),  
    CI_high = exp(Estimate + 1.96 * `Std. Error`)  
  ) %>%  
  select(  
    term,  
    Estimate,  
    SE = `Std. Error`,  
    OR,  
    CI_low,  
    CI_high  
  )
```

lrt_table

```
lrt_table <- as.data.frame(
  model_binom_lrt_controls$anova_table
) %>%
  tibble::rownames_to_column("effect") %>%
  transmute(
    effect,
    LR_Chisq = Chisq,
    p = `Pr(>Chisq)`
  )
```

name comparison

```
coef_table$term
```

```
[1] "(Intercept)"      "InfoDistribution1" "InfoImportance1"
[4] "Age_z"            "sex"              "SES_z"
[7] "hispanic"         "NFC_z"            "Narcissism_z"
[10] "Psychopathy_z"   "Machiavellianism_z" "SVO_z"
```

```
lrt_table$effect
```

```
[1] "InfoDistribution" "InfoImportance"   "Age_z"
[4] "sex"              "SES_z"            "hispanic"
[7] "NFC_z"            "Narcissism_z"     "Psychopathy_z"
[10] "Machiavellianism_z" "SVO_z"
```

term_map

```
term_map <- tibble::tribble(
  ~term, ~effect, ~Predictor,

  "(Intercept)",
  NA_character_,
  "Intercept",

  "InfoDistribution1",
  "InfoDistribution",
  "Information Distribution",
```

```
"InfoImportance1",
"InfoImportance",
"Information Importance",

"Age_z",
"Age_z",
"Age",

"sex",
"sex",
"Sex (1 = Female)",

"SES_z",
"SES_z",
"Socioeconomic Status",

"hispanic",
"hispanic",
"Hispanic",

"NFC_z",
"NFC_z",
"Need for Cognition",

"Narcissism_z",
"Narcissism_z",
"Narcissism",

"Psychopathy_z",
"Psychopathy_z",
"Psychopathy",

"Machiavellianism_z",
"Machiavellianism_z",
"Machiavellianism",

"SVO_z",
"SVO_z",
"Social Value Orientation"
)
```

left join

```
table_control_model <- term_map %>%  
  left_join(coef_table, by = "term") %>%  
  left_join(lrt_table, by = "effect")
```

table data

```
table_control_model <- table_control_model %>%  
  mutate(  
    Estimate = sprintf("%.2f", Estimate),  
    SE = sprintf("%.2f", SE),  
  
    LR_Chisq = case_when(  
      Predictor == "Intercept" ~ "-",  
      is.na(LR_Chisq) ~ "",  
      TRUE ~ sprintf("%.2f", LR_Chisq)  
    ),  
  
    p = case_when(  
      Predictor == "Intercept" ~ "-",  
      is.na(p) ~ "",  
      p < .001 ~ "< .001",  
      TRUE ~ sub("^0", "", sprintf("%.3f", p))  
    ),  
  
    OR_CI = sprintf(  
      "%.2f [%.2f, %.2f]",  
      OR,  
      CI_low,  
      CI_high  
    )  
  ) %>%  
  select(  
    Predictor,  
    Estimate,  
    SE,  
    LR_Chisq,  
    p,  
    OR_CI  
  )
```

```

table_control_model <- bind_rows(
  table_control_model %>% slice(1),

  tibble(
    Predictor = "Item Characteristics",
    Estimate = "",
    SE = "",
    LR_Chisq = "",
    p = "",
    OR_CI = ""
  ),

  table_control_model %>% slice(2:3),

  tibble(
    Predictor = "Candidate Control Variables",
    Estimate = "",
    SE = "",
    LR_Chisq = "",
    p = "",
    OR_CI = ""
  ),

  table_control_model %>% slice(4:n())
)

```

```
table_control_model
```

```
# A tibble: 14 x 6
```

	Predictor <chr>	Estimate <chr>	SE <chr>	LR_Chisq <chr>	p <chr>	OR_CI <chr>
1	Intercept	"0.86"	"0.26"	"-"	"-"	"2.37 [1.42, 3.9~
2	Item Characteristics	"	"	"	"	"
3	Information Distribution	"0.22"	"0.11"	"3.34"	".068"	"1.24 [1.00, 1.5~
4	Information Importance	"-0.45"	"0.11"	"10.28"	".001"	"0.64 [0.51, 0.8~
5	Candidate Control Variables	"	"	"	"	"
6	Age	"-0.10"	"0.11"	"0.82"	".364"	"0.91 [0.73, 1.1~
7	Sex (1 = Female)	"-0.59"	"0.24"	"5.80"	".016"	"0.56 [0.35, 0.8~
8	Socioeconomic Status	"0.06"	"0.10"	"0.39"	".530"	"1.07 [0.87, 1.3~
9	Hispanic	"-0.05"	"0.21"	"0.07"	".797"	"0.95 [0.63, 1.4~
10	Need for Cognition	"0.32"	"0.10"	"10.02"	".002"	"1.38 [1.13, 1.6~
11	Narcissism	"-0.15"	"0.13"	"1.25"	".263"	"0.86 [0.66, 1.1~

12 Psychopathy	"-0.08"	"0.11"	"0.47"	".494"	"0.93	[0.75, 1.1~
13 Machiavellianism	"-0.12"	"0.13"	"0.80"	".372"	"0.89	[0.68, 1.1~
14 Social Value Orientation	"-0.00"	"0.12"	"0.00"	".988"	"1.00	[0.79, 1.2~

full model

coef_table

```
coef_table <- as.data.frame(
  summary(model_binom_lrt$full_model)$coefficients
) %>%
  tibble::rownames_to_column("term") %>%
  mutate(
    OR = exp(Estimate),
    CI_low = exp(Estimate - 1.96 * `Std. Error`),
    CI_high = exp(Estimate + 1.96 * `Std. Error`)
  ) %>%
  select(
    term,
    Estimate,
    SE = `Std. Error`,
    OR,
    CI_low,
    CI_high
  )
```

lrt_table

```
lrt_table <- as.data.frame(
  model_binom_lrt$anova_table
) %>%
  tibble::rownames_to_column("effect") %>%
  transmute(
    effect,
    LR_Chisq = Chisq,
    p = `Pr(>Chisq)`
  )
```

name comparison

```
coef_table$term
```

```
[1] "(Intercept)"
[2] "structure_number011"
[3] "motivation_number011"
[4] "InfoDistribution1"
[5] "InfoImportance1"
[6] "sex"
[7] "NeedCog_z"
[8] "structure_number011:motivation_number011"
[9] "structure_number011:InfoDistribution1"
[10] "motivation_number011:InfoDistribution1"
[11] "structure_number011:InfoImportance1"
[12] "motivation_number011:InfoImportance1"
[13] "InfoDistribution1:InfoImportance1"
[14] "structure_number011:motivation_number011:InfoDistribution1"
[15] "structure_number011:motivation_number011:InfoImportance1"
[16] "structure_number011:InfoDistribution1:InfoImportance1"
[17] "motivation_number011:InfoDistribution1:InfoImportance1"
[18] "structure_number011:motivation_number011:InfoDistribution1:InfoImportance1"
```

```
lrt_table$effect
```

```
[1] "structure_number01"
[2] "motivation_number01"
[3] "InfoDistribution"
[4] "InfoImportance"
[5] "sex"
[6] "NeedCog_z"
[7] "structure_number01:motivation_number01"
[8] "structure_number01:InfoDistribution"
[9] "motivation_number01:InfoDistribution"
[10] "structure_number01:InfoImportance"
[11] "motivation_number01:InfoImportance"
[12] "InfoDistribution:InfoImportance"
[13] "structure_number01:motivation_number01:InfoDistribution"
[14] "structure_number01:motivation_number01:InfoImportance"
[15] "structure_number01:InfoDistribution:InfoImportance"
[16] "motivation_number01:InfoDistribution:InfoImportance"
[17] "structure_number01:motivation_number01:InfoDistribution:InfoImportance"
```

term_map

```
term_map <- tibble::tribble(  
  ~term, ~effect, ~Predictor,  
  
  "(Intercept)", NA_character_, "Intercept",  
  
  "sex",  
  "sex",  
  "Sex (1 = Female)",  
  
  "NeedCog_z",  
  "NeedCog_z",  
  "Need for Cognition",  
  
  "structure_number011",  
  "structure_number01",  
  "Interindividual (0) vs. Intergroup (1)",  
  
  "motivation_number011",  
  "motivation_number01",  
  "Proself (0) vs. Prosocial (1)",  
  
  "InfoDistribution1",  
  "InfoDistribution",  
  "Common (0) vs. Unique (1)",  
  
  "InfoImportance1",  
  "InfoImportance",  
  "Low (0) vs. High (1)",  
  
  "structure_number011:motivation_number011",  
  "structure_number01:motivation_number01",  
  "Structure × Motivation",  
  
  "structure_number011:InfoDistribution1",  
  "structure_number01:InfoDistribution",  
  "Structure × Distribution",  
  
  "motivation_number011:InfoDistribution1",  
  "motivation_number01:InfoDistribution",  
  "Motivation × Distribution",
```

```

"structure_number011:InfoImportance1",
"structure_number01:InfoImportance",
"Structure × Importance",

"motivation_number011:InfoImportance1",
"motivation_number01:InfoImportance",
"Motivation × Importance",

"InfoDistribution1:InfoImportance1",
"InfoDistribution:InfoImportance",
"Distribution × Importance",

"structure_number011:motivation_number011:InfoDistribution1",
"structure_number01:motivation_number01:InfoDistribution",
"Structure × Motivation × Distribution",

"structure_number011:motivation_number011:InfoImportance1",
"structure_number01:motivation_number01:InfoImportance",
"Structure × Motivation × Importance",

"structure_number011:InfoDistribution1:InfoImportance1",
"structure_number01:InfoDistribution:InfoImportance",
"Structure × Distribution × Importance",

"motivation_number011:InfoDistribution1:InfoImportance1",
"motivation_number01:InfoDistribution:InfoImportance",
"Motivation × Distribution × Importance",

"structure_number011:motivation_number011:InfoDistribution1:InfoImportance1",
"structure_number01:motivation_number01:InfoDistribution:InfoImportance",
"Structure × Motivation × Distribution × Importance"
)

```

left join

```

table_full_model <- term_map %>%
  left_join(coef_table, by = "term") %>%
  left_join(lrt_table, by = "effect")

```

table data

```

table_full_model <- table_full_model %>%
  mutate(
    Estimate = sprintf("%.2f", Estimate),
    SE = sprintf("%.2f", SE),

    LR_Chisq = case_when(
      Predictor == "Intercept" ~ "-",
      is.na(LR_Chisq) ~ "",
      TRUE ~ sprintf("%.2f", LR_Chisq)
    ),

    p = case_when(
      Predictor == "Intercept" ~ "-",
      is.na(p) ~ "",
      p < .001 ~ "< .001",
      TRUE ~ sub("^0", "", sprintf("%.3f", p))
    ),

    OR_CI = sprintf(
      "%.2f [%.2f, %.2f]",
      OR,
      CI_low,
      CI_high
    )
  ) %>%
  select(
    Predictor,
    Estimate,
    SE,
    LR_Chisq,
    p,
    OR_CI
  )

```

```

table_full_model <- bind_rows(
  table_full_model %>% slice(1),

  tibble(
    Predictor = "Control Variables",
    Estimate = "",
    SE = "",
    LR_Chisq = "",

```

```

      p = "",
      OR_CI = ""
    ),
    table_full_model %>% slice(2:3),

    tibble(
      Predictor = "Between Manipulations",
      Estimate = "",
      SE = "",
      LR_Chisq = "",
      p = "",
      OR_CI = ""
    ),
    table_full_model %>% slice(4:5),

    tibble(
      Predictor = "Within Manipulations",
      Estimate = "",
      SE = "",
      LR_Chisq = "",
      p = "",
      OR_CI = ""
    ),
    table_full_model %>% slice(6:7),

    tibble(
      Predictor = "Interactions",
      Estimate = "",
      SE = "",
      LR_Chisq = "",
      p = "",
      OR_CI = ""
    ),
    table_full_model %>% slice(8:n())
  )
)

```

```
table_full_model
```

```
# A tibble: 22 x 6
```

	Predictor	Estimate	SE	LR_Chisq	p	OR_CI
	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
1	Intercept	"0.79"	"0.20"	"-"	"-"	"2.2~"

2 Control Variables	"	"	"	"	"
3 Sex (1 = Female)	"-0.58"	"0.21"	"7.25"	".007"	"0.5~
4 Need for Cognition	"0.38"	"0.10"	"15.04"	"< .00~	"1.4~
5 Between Manipulations	"	"	"	"	"
6 Interindividual (0) vs. Intergroup (1)	"-0.05"	"0.09"	"0.30"	".586"	"0.9~
7 Proself (0) vs. Prosocial (1)	"-0.38"	"0.09"	"15.29"	"< .00~	"0.6~
8 Within Manipulations	"	"	"	"	"
9 Common (0) vs. Unique (1)	"0.22"	"0.11"	"3.32"	".068"	"1.2~
10 Low (0) vs. High (1)	"-0.46"	"0.11"	"10.47"	".001"	"0.6~
# i 12 more rows					

```
rm(term_map)
```

panels

```
extract_panel <- function(model, clue_label) {

  as.data.frame(model$anova_table) %>%
    tibble::rownames_to_column("effect") %>%
    filter(
      effect %in% c(
        "structure_number01",
        "motivation_number01",
        "structure_number01:motivation_number01"
      )
    ) %>%
    transmute(
      Clue = clue_label,
      effect,
      LR_Chisq = Chisq,
      p = `Pr(>Chisq)`
    )
}
```

```
panel_results <- bind_rows(
  extract_panel(lrt_high_common, "Common, High"),
  extract_panel(lrt_high_unique, "Unique, High"),
  extract_panel(lrt_low_common, "Common, Low"),
  extract_panel(lrt_low_unique, "Unique, Low")
)
```

```

panel_results <- panel_results %>%
  mutate(
    result = paste0(
      format_stat(LR_Chisq),
      " (",
      format_stat(p, p_value = TRUE),
      ")"
    ),
    effect = recode(
      effect,
      "structure_number01" = "Structure",
      "motivation_number01" = "Motivation",
      "structure_number01:motivation_number01" =
        "Structure × Motivation"
    )
  )

```

table_panel

```

table_panel <- panel_results %>%
  select(
    Clue,
    effect,
    result
  ) %>%
  tidyr::pivot_wider(
    names_from = effect,
    values_from = result
  )

```

table_panel

```

# A tibble: 4 x 4
  Clue          Structure Motivation `Structure × Motivation`
<chr>         <chr>      <chr>      <chr>
1 Common, High 0.00 (1.00) 10.39 (.001**) 14.51 (< .001**)
2 Unique, High 7.83 (.01**) 8.57 (.003**) 1.01 (.31)
3 Common, Low 0.91 (.34) 1.04 (.31) 1.35 (.25)
4 Unique, Low 0.39 (.53) 3.57 (.06) 1.51 (.22)

```

```
rm(panel_results, extract_panel)
```

estimated probabilities

dataframe

```
probability_table_data <- bind_rows(  
  as.data.frame(lc_emm) %>% mutate(Clue = "Common, Low"),  
  as.data.frame(hc_emm) %>% mutate(Clue = "Common, High"),  
  as.data.frame(lu_emm) %>% mutate(Clue = "Unique, Low"),  
  as.data.frame(hu_emm) %>% mutate(Clue = "Unique, High")  
) %>%  
  mutate(  
    P_CI = format_p_ci(  
      prob,  
      asymp.LCL,  
      asymp.UCL  
    ),  
  
    Condition = paste(  
      structure_number01,  
      motivation_number01,  
      sep = ", "  
    )  
  )  
)
```

```
table_probability <- probability_table_data %>%  
  select(  
    Clue,  
    Condition,  
    P_CI  
  ) %>%  
  tidyr::pivot_wider(  
    names_from = Condition,  
    values_from = P_CI  
  )
```

```
table_probability <- table_probability %>%  
  mutate(  
    Clue = factor(  
      Clue,  

```

```

      levels = c(
        "Common, Low",
        "Common, High",
        "Unique, Low",
        "Unique, High"
      )
    )
  ) %>%
  arrange(Clue) %>%
  select(
    Clue,
    `Interindividual, Prosocial`,
    `Interindividual, Proself`,
    `Intergroup, Prosocial`,
    `Intergroup, Proself`
  )

```

table_probability

```

# A tibble: 4 x 5
  Clue      Interindividual, Pro~1 Interindividual, Pro~2 Intergroup, Prosocial~3
<fct>      <chr>                  <chr>                  <chr>
1 Common, ~ 0.80 [0.52, 0.93]    0.81 [0.54, 0.94]    0.93 [0.75, 0.98]
2 Common, ~ 0.91 [0.75, 0.97]    0.24 [0.10, 0.48]    0.60 [0.35, 0.81]
3 Unique, ~ 0.75 [0.55, 0.88]    0.52 [0.31, 0.73]    0.72 [0.50, 0.86]
4 Unique, ~ 0.79 [0.59, 0.91]    0.42 [0.22, 0.64]    0.43 [0.23, 0.65]
# i abbreviated names: 1: `Interindividual, Prosocial`,
#   2: `Interindividual, Proself`, 3: `Intergroup, Prosocial`
# i 1 more variable: `Intergroup, Proself` <chr>

```

follow-up comparisons

dataframe

```

hc_structure_or <- summary(
  pairs(
    hc_structure,
    reverse = TRUE,
    adjust = "none"
  ),
  type = "response",

```

```

    infer = c(TRUE, FALSE)
  ) %>%
    as.data.frame()

hc_motivation_or <- summary(
  pairs(
    hc_motivation,
    adjust = "none"
  ),
  type = "response",
  infer = c(TRUE, FALSE)
) %>%
  as.data.frame()

hu_structure_or <- summary(
  pairs(
    hu_structure,
    reverse = TRUE,
    adjust = "none"
  ),
  type = "response",
  infer = c(TRUE, FALSE)
) %>%
  as.data.frame()

hu_motivation_or <- summary(
  pairs(
    hu_motivation,
    adjust = "none"
  ),
  type = "response",
  infer = c(TRUE, FALSE)
) %>%
  as.data.frame()

```

```

hc_lrt_p <- c(
  lrt_hc_structure_proself$anova_table[
    "structure_number01", "Pr(>Chisq)"
  ],

  lrt_hc_structure_prosocial$anova_table[
    "structure_number01", "Pr(>Chisq)"
  ]
)

```

```

],

lrt_hc_motivation_interindividual$anova_table[
  "motivation_number01", "Pr(>Chisq)"
],

lrt_hc_motivation_intergroup$anova_table[
  "motivation_number01", "Pr(>Chisq)"
]
)

```

```

hc_lrt_p_holm <- p.adjust(
  hc_lrt_p,
  method = "holm"
)

```

```

hu_structure_p <- lrt_high_unique$anova_table[
  "structure_number01",
  "Pr(>Chisq)"
]

hu_motivation_p <- lrt_high_unique$anova_table[
  "motivation_number01",
  "Pr(>Chisq)"
]

```

```

common_high_table <- tibble(
  Clue = c(
    "Common, High",
    "",
    "",
    ""
  ),

  Comparison = c(
    "Proself: Interindividual vs. Intergroup",
    "Prosocial: Interindividual vs. Intergroup",
    "Interindividual: Proself vs. Prosocial",
    "Intergroup: Proself vs. Prosocial"
  ),

  OR_CI = c(

```

```

format_or_ci(
  hc_structure_or$odds.ratio[
    hc_structure_or$motivation_number01 == "Proself"
  ],
  hc_structure_or$asympt.LCL[
    hc_structure_or$motivation_number01 == "Proself"
  ],
  hc_structure_or$asympt.UCL[
    hc_structure_or$motivation_number01 == "Proself"
  ]
),

format_or_ci(
  hc_structure_or$odds.ratio[
    hc_structure_or$motivation_number01 == "Prosocial"
  ],
  hc_structure_or$asympt.LCL[
    hc_structure_or$motivation_number01 == "Prosocial"
  ],
  hc_structure_or$asympt.UCL[
    hc_structure_or$motivation_number01 == "Prosocial"
  ]
),

format_or_ci(
  hc_motivation_or$odds.ratio[
    hc_motivation_or$structure_number01 == "Interindividual"
  ],
  hc_motivation_or$asympt.LCL[
    hc_motivation_or$structure_number01 == "Interindividual"
  ],
  hc_motivation_or$asympt.UCL[
    hc_motivation_or$structure_number01 == "Interindividual"
  ]
),

format_or_ci(
  hc_motivation_or$odds.ratio[
    hc_motivation_or$structure_number01 == "Intergroup"
  ],
  hc_motivation_or$asympt.LCL[
    hc_motivation_or$structure_number01 == "Intergroup"
  ]
)

```

```

    ],
    hc_motivation_or$asyp.UCL[
      hc_motivation_or$structure_number01 == "Intergroup"
    ]
  )
),

p = format_stat(
  hc_lrt_p_holm,
  p_value = TRUE
)
)

```

```

unique_high_table <- tibble(
  Clue = c(
    "Unique, High",
    ""
  ),

  Comparison = c(
    "Interindividual vs. Intergroup",
    "Proself vs. Prosocial"
  ),

  OR_CI = c(
    format_or_ci(
      hu_structure_or$odds.ratio,
      hu_structure_or$asyp.LCL,
      hu_structure_or$asyp.UCL
    ),

    format_or_ci(
      hu_motivation_or$odds.ratio,
      hu_motivation_or$asyp.LCL,
      hu_motivation_or$asyp.UCL
    )
  ),

  p = c(
    format_stat(
      hu_structure_p,
      p_value = TRUE
    )
  )
)

```

```

    ),
    format_stat(
      hu_motivation_p,
      p_value = TRUE
    )
  )
)

```

```

table_followup <- bind_rows(
  common_high_table,
  unique_high_table
)

table_followup

```

```

# A tibble: 6 x 4
  Clue          Comparison OR_CI          p
<chr>         <chr>         <chr>         <chr>
1 "Common, High" Prosself: Interindividual vs. Intergroup 0.16 [0.04, 0.~ .05
2 ""           Prosocial: Interindividual vs. Intergroup 6.30 [1.63, 24~ .02*
3 ""           Interindividual: Prosself vs. Prosocial 0.03 [0.01, 0.~ < .0~
4 ""           Intergroup: Prosself vs. Prosocial 1.30 [0.34, 4.~ .74
5 "Unique, High" Interindividual vs. Intergroup 3.32 [1.39, 7.~ .01**
6 ""           Prosself vs. Prosocial 0.28 [0.12, 0.~ .003~

```

```
rm(unique_high_table, common_high_table, hu_motivation_p, hu_structure_p, hc_lrt_p_holm, hc_
```

figures

```

emm_sm_df <- bind_rows(
  as.data.frame(lc_emm) %>%
    mutate(
      InfoDistribution = "Common",
      InfoImportance = "Low"
    ),
  as.data.frame(hc_emm) %>%
    mutate(

```

```

      InfoDistribution = "Common",
      InfoImportance = "High"
    ),

as.data.frame(lu_emm) %>%
  mutate(
    InfoDistribution = "Unique",
    InfoImportance = "Low"
  ),

as.data.frame(hu_emm) %>%
  mutate(
    InfoDistribution = "Unique",
    InfoImportance = "High"
  )
) %>%
mutate(
  structure_number01 = factor(
    structure_number01,
    levels = c("Interindividual", "Intergroup")
  ),
  motivation_number01 = factor(
    motivation_number01,
    levels = c("Proself", "Prosocial")
  ),
  InfoDistribution = factor(
    InfoDistribution,
    levels = c("Common", "Unique")
  ),
  InfoImportance = factor(
    InfoImportance,
    levels = c("Low", "High")
  )
)

```

```

fig_emm_sm <- ggplot(
  emm_sm_df,
  aes(
    x = structure_number01,
    y = prob,
    fill = motivation_number01
  )
)

```

```

) +
  geom_col(
    position = position_dodge(width = 0.8),
    width = 0.7,
    alpha = 0.5,
    color = "black",
    linewidth = 0.2
  ) +
  geom_errorbar(
    aes(
      ymin = asymp.LCL,
      ymax = asymp.UCL
    ),
    position = position_dodge(width = 0.8),
    width = 0.25,
    linewidth = 0.8
  ) +
  facet_grid(
    InfoDistribution ~ InfoImportance
  ) +
  scale_y_continuous(
    breaks = seq(0, 1, 0.2),
    limits = c(0, 1)
  ) +
  theme_apache() +
  theme(
    text = element_text(
      size = 12,
      family = "serif",
      color = "black"
    ),
    axis.title = element_text(
      size = 12,
      face = "plain",
      color = "black"
    ),
    axis.text = element_text(
      size = 11,
      face = "plain",
      color = "black"
    ),
    strip.text = element_text(

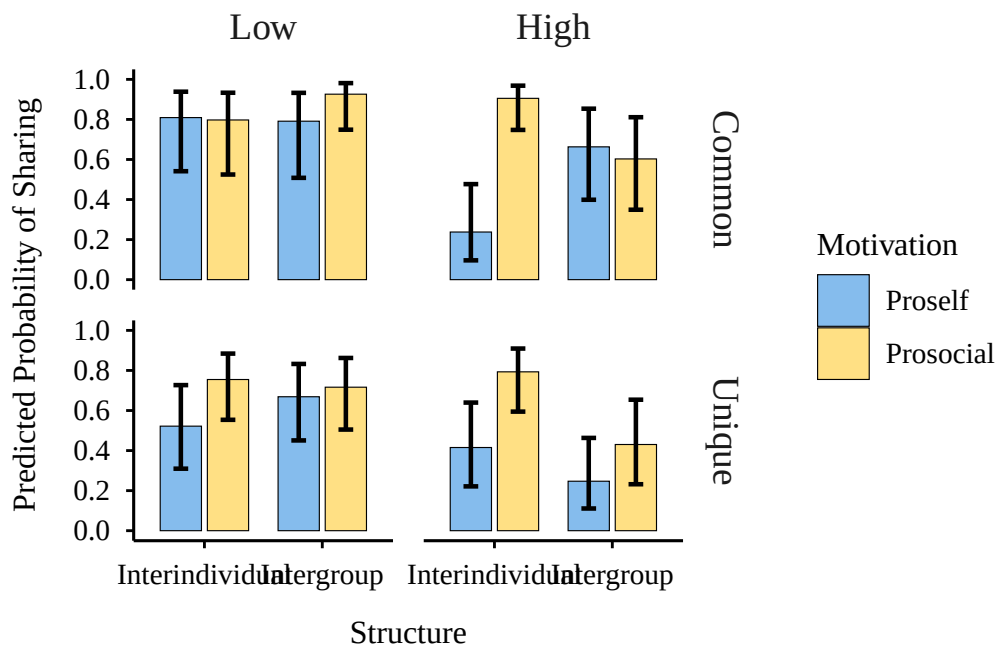
```

```

    size = 12,
    face = "plain",
    family = "serif"
  ),
  legend.title = element_text(
    size = 12,
    face = "plain"
  ),
  legend.text = element_text(
    size = 11,
    face = "plain"
  ),
  axis.ticks = element_line(
    color = "black",
    linewidth = 0.5
  ),
  axis.ticks.length = unit(0.10, "cm"),
  axis.line = element_line(
    color = "black"
  ),
  panel.grid.major = element_blank(),
  panel.grid.minor = element_blank(),
  panel.border = element_blank(),
  panel.spacing = unit(0.8, "lines")
) +
labs(
  x = "Structure",
  y = "Predicted Probability of Sharing",
  fill = "Motivation"
) +
scale_fill_manual(
  name = "Motivation",
  values = c(
    "Proself" = "#0C7BDC",
    "Prosocial" = "#FFC20A"
  )
)
)

fig_emm_sm

```



save

```
# Create workbook to save the analyses
wb <- createWorkbook()
excel_export_file_location <- "../artifacts/binomialLRT_output.xlsx"

# Start fresh each time
if (file.exists(excel_export_file_location)) {
  file.remove(excel_export_file_location)
}
```

[1] TRUE

```
save_output_to_workbook(
  wb,
  sheet_name = "control model",
  output = table_control_model
)

save_output_to_workbook(
```

```

wb,
  sheet_name = "full model",
  output = table_full_model
)

save_figure_to_workbook(
  wb = wb,
  sheet_name = "Figure",
  figure_file = "../figures/EstProb.png",
  plot_obj = fig_emm_sm,
  width = 8,
  height = 6,
  dpi = 300
)

save_output_to_workbook(
  wb,
  sheet_name = "Panels",
  output = table_panel
)

save_output_to_workbook(
  wb,
  sheet_name = "Probability",
  output = table_probability
)

save_output_to_workbook(
  wb,
  sheet_name = "Follow up",
  output = table_followup
)

```

export

```

data.frame(
  Sheet = seq_along(names(wb)),
  Worksheet = names(wb)
)

```

	Sheet	Worksheet
1	1	control model
2	2	full model
3	3	Figure
4	4	Panels
5	5	Probability
6	6	Follow up

```
cat("Formatting worksheets...\n")
```

Formatting worksheets...

```
for (sheet in names(wb)) {  
  
  cat("Formatting:", sheet, "\n")  
  
  setColWidths(  
    wb,  
    sheet = sheet,  
    cols = 1:30,  
    widths = "auto"  
  )  
}
```

Formatting: control model
Formatting: full model
Formatting: Figure
Formatting: Panels
Formatting: Probability
Formatting: Follow up

```
cat("Finished formatting worksheets.\n")
```

Finished formatting worksheets.

```
cat("Saving workbook...\n")
```

Saving workbook...

```
saveWorkbook(  
  wb,  
  excel_export_file_location,  
  overwrite = TRUE  
)  
  
cat("Workbook saved successfully.\n")
```

Workbook saved successfully.

end _____

```
print("end of script")
```

```
[1] "end of script"
```